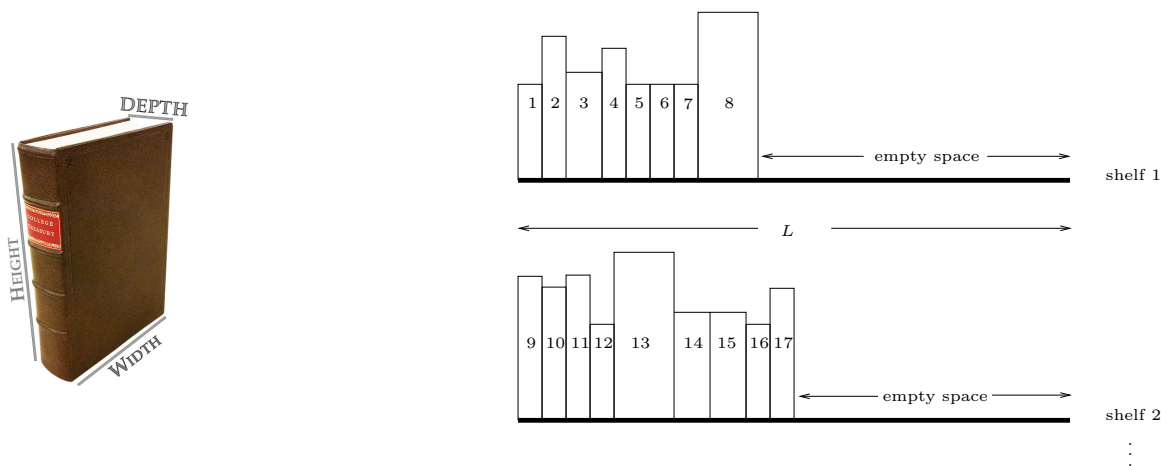


Homework Assignment #1
Due: September 11, 2024, by 11:59 pm

- **You must submit your assignment through the Crowdmark system.** You will receive by email an invitation through which you can submit your work in the form of separate PDF documents with your answers to each question of the assignment. To work with a partner, you and your partner must form a group on Crowdmark. Crowdmark does not enforce a limit on the size of groups. **The course policy that limits the size of each group to at most two remains in effect:** submissions by groups of more than two persons will not be graded.
- It is your responsibility to ensure that the PDF files you submit are legible. To this end, I encourage you to learn and use the LaTeX typesetting system, which is designed to produce high-quality documents that contain mathematical notation. You are not required to produce the PDF files you submit using LaTeX; you may produce it any way you wish, as long as the resulting document is legible.
- By virtue of submitting this assignment you (and your partner, if you have one) acknowledge that you are aware of the policy on homework collaboration for this course.^a
- For any question, you may use data structures and algorithms previously described in class, or in prerequisites of this course, without describing them. You may also use any result that we covered in class, or is in the assigned sections of the official course textbooks, by referring to it.
- Unless we explicitly state otherwise, you may describe algorithms in high-level pseudocode or point-form English, whichever leads to a clearer and simpler description. Do not provide executable code.
- Unless we explicitly state otherwise, you should justify your answers. Your paper will be graded based on **the correctness and efficiency of your answers, and the clarity, precision, and conciseness of your presentation.**

^a“In each homework assignment you may collaborate with at most one other student who is currently taking CSCC73. If you collaborate with another student on an assignment, you and your partner must submit only one copy of your solution, with both of your names. The solution will be graded in the usual way and both partners will receive the same mark. Collaboration involving more than two students is not allowed. **For help with your homework you may consult only the instructor, TAs, your homework partner (if you have one), your textbook, and your class notes. You may not consult any other source.**”

Question 1. (15 marks) We are given a sequence of n books of depths $d_1, d_2, \dots, d_n$ (see figure on the left below). We must place these books one after the other **in the given order** on successive shelves, each of length $L \geq \max(d_1, d_2, \dots, d_n)$, so that the sum of the depths of the books on each shelf does not exceed L .



The following greedy algorithm places books on the shelves according to these rules. Intuitively it packs as many books as possible on each shelf, and uses a new shelf if and only if the next book does not fit on the current shelf.

```

 $i := 1$       ► next book to place on a shelf
 $s := 1$       ► current shelf
 $t := 0$       ► total depth of books on current shelf
while  $i \leq n$  do
  if  $t + d_i \leq L$  then    ► the next book fits on the current shelf
    if  $t = 0$  then place book  $i$  in the leftmost spot of shelf  $s$ 
    else place book  $i$  after book  $i - 1$  on shelf  $s$ 
     $t := t + d_i$ ;  $i := i + 1$ 
  else
    ► the next book does not fit on the current shelf
     $s := s + 1$ ;  $t := 0$ 

```

- a. Does this algorithm minimize the *sum of empty spaces on the shelves used*? Justify your answer.
- b. Does this algorithm minimize the *maximum empty space on any shelf used*? Justify your answer.

Note: When disproving the correctness of a proposed algorithm, it is not sufficient to just give a counterexample, i.e., an input on which the algorithm does not work correctly. You must also indicate what the algorithm does on your counterexample, and why that is not the correct answer. Your counterexamples should be short and illuminating.

Question 2. (20 marks) Let $\mathcal{I}$ be a set of intervals over the real numbers. (For specificity say these are closed intervals, but this is not important.) A subset $\mathcal{C}$ of $\mathcal{I}$ is a **cover of $\mathcal{I}$** if for every interval $I \in \mathcal{I}$ there is an interval $I' \in \mathcal{C}$ such that I and I' intersect (even at one point). We wish to find a **minimum cardinality** cover of $\mathcal{I}$ — i.e., a cover of $\mathcal{I}$ with as few intervals as possible.

- a. True or false? For every set of intervals $\mathcal{I}$ and every minimum cardinality cover $\mathcal{C}$ of $\mathcal{I}$, no two intervals in $\mathcal{C}$ intersect. Justify your answer.
- b. Describe an $O(n^2)$ greedy algorithm that, given a set $\mathcal{I}$ of n intervals, returns a minimum cardinality cover of $\mathcal{I}$. Prove the correctness of your algorithm and analyze its running time.